Nivela 3D

BONUS · NIVELA 3D

Anti-Waste Checklist

The checks that prevent failures and throw less filament, time and profit in the trash. Print it and keep it next to your printer.

Exclusive bonus for Nivela 3D customers.

PART 01

What a failure really costs

Every lost print takes material, electricity, machine hours and often the customer's deadline with it.

If 1 in 10 pieces fails, the other 9 have to pay for it. That's why Nivela 3D has a **failure rate** field: it spreads that cost across the price. But the best path is to **lower the rate**. Every point less is straight profit.

How to measure: for a month, write down how many prints you started and how many were thrown away or redone. Failures ÷ total × 100 = your rate. Update that number in Costs & channels.

PART 02

Before slicing

☐ The model has no errors (closed mesh, no holes). If the slicer flags a problem, fix it first.

☐ The chosen orientation has the largest flat face on the bed.

☐ The piece is rotated to need as little support as possible.

☐ Details that need to look good face up or sideways, not resting on supports.

☐ Large pieces are split into parts when that lowers risk and support.

☐ I confirmed the dimensions with the customer before printing (especially functional parts).

PART 03

In the slicer

☐ Correct filament profile (nozzle and bed temperatures recommended by the manufacturer).

☐ Infill suited to the use: a decorative piece doesn't need high infill.

☐ Wall count chosen for strength, instead of raising infill.

☐ Sensible layer height: thicker layers print faster when detail doesn't matter.

☐ Supports only where needed (painted on or with an adjusted overhang angle).

☐ Brim on tall, thin pieces or ones with little contact area.

☐ I checked the preview layer by layer, especially the first layer and bridges.

☐ I noted the estimated weight and time to enter in the calculator.

PART 04

Before hitting print

☐ Clean bed (no finger grease) with the right adhesion for the filament.

☐ Bed leveling or calibration up to date.

☐ Clean nozzle, no stuck filament.

☐ Dry filament. If it pops, strings or bubbles, it's wet and needs drying.

☐ Spool turns freely, no tangles, filament not caught on the side.

☐ Enough filament for the whole piece (weigh the spool or check what's left).

☐ Right file, on the right machine, with the right profile.

PART 05

First layer and first minutes

Most failures show up right at the start. Watching the first 10 minutes saves hours.

☐ The first layer is even, well squished and has no gaps.

☐ The edges aren't lifting (warping).

☐ No strings dragging and the nozzle isn't hitting the piece.

☐ If something's wrong, cancel now. Restarting after 5 minutes costs less than losing 6 hours.

PART 06

During the print

☐ No drafts or sudden temperature changes in the room.

☐ Camera or regular check-ins on long prints.

☐ For overnight prints: piece tested before, enough filament and first layer checked.

☐ If it fails midway, write down the cause (adhesion, clog, filament, power…).

PART 07

After printing

☐ Let the bed cool before removing the piece (fewer marks and fewer broken parts).

☐ Remove supports with the right tool, without forcing the piece.

☐ Log the real finishing time in Nivela 3D so the price reflects reality.

☐ Keep purge scraps and defective pieces separated by material.

☐ Pieces with minor defects can become samples, freebies or display pieces.

PART 08

Filament storage

☐ Open spools stored in a sealed box or bag with silica gel.

☐ Silica replaced or regenerated when it changes color (if it's the indicating kind).

☐ Spool labeled with material, color and opening date.

☐ Filament end clipped into the spool hole so it doesn't unwind and tangle.

☐ Moisture-sensitive materials (like PETG, TPU and nylon) dried before use.

PART 09

Maintenance that prevents failures

How often	What to do
Every print	Clean the bed and check the first layer.
Every week	Clean the nozzle, check belts and loose screws, remove dust and filament scraps.
Every month	Lubricate the axes per the manual, recheck leveling and extrusion.
When it shows signs	Replace the nozzle (wear, recurring clogs) and the bed surface (loss of adhesion).

Tip: wear parts like the nozzle and bed surface are built into the piece cost through the printer wear field in Nivela 3D. So when it's time to replace them, the money already came from sales.

PART 10

Monthly failure log

Date	Piece	What happened	Loss (g / h)